Answer Key - Mathematics - K-12 Level Assessment 12 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** A
   Options: A) $10.00; B) $10.50; C) $13.50; D) $16.00
   Note: Correct option: A - $10.00

2. **Answer:** B
   Options: A) 18; B) 24; C) 25; D) 29
   Note: Correct option: B - 24

3. **Answer:** A
   Options: A) 18 x+15; B) 18 x+3; C) 7 x+8; D) 2 x+8
   Note: Correct option: A - 18 x+15

4. **Answer:** C
   Options: A) 156; B) 110; C) 94; D) 48
   Note: Correct option: C - 94

5. **Answer:** C
   Options: A) 0.25 1.6 1.0; B) 1.0 0.25 1.6; C) 0.25 1.0 1.6; D) 1.6 1.0 0.25
   Note: Correct option: C - 0.25 1.0 1.6

True / False

6. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: '4'.

7. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

8. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: '12.68'.

9. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: '13 minutes'.

10. **Answer:** True
    Options: A) True; B) False
    Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** Since $\mathbf{v}_n$ is always a projection onto $\mathbf{v}_0$,

$$\mathbf{v}_n = a_n \mathbf{v}_0$$

for some constant a_n. Similarly,

$$\mathbf{w}_n = b_n \mathbf{w}_0$$

for some constant b_n. [asy] unitsize(1.5 cm); pair[] V, W; V[0] = (1,3); W[0] = (4,0); V[1] = (W[0] + reflect((0,0),V[0])*(W[0]))/2; W[1] = (V[1] + reflect((0,0),W[0])*(V[1]))/2; V[2] = (W[1] + reflect((0,0),V[0])*(W[1]))/2; W[2] = (V[2] + reflect((0,0),W[0])*(V[2]))/2; V[3] = (W[2] + reflect((0,0),V[0])*(W[2]))/2; W[3] = (V[3] + reflect((0,0),W[0])*(V[3]))/2; draw((-1,0)–(5,0)); draw((0,-1)–(0,4)); draw((0,0)–V[0],red,Arrow(6)); draw((0,0)–W[0],red,Arrow(6)); draw((0,0)–V[1],red,Arrow(6)); draw((0,0)–W[1],red,Arrow(6)); draw((0,0)–V[2],red,Arrow(6)); draw((0,0)–W[2],red,Arrow(6)); draw(W[0]–V[1]–W[1]–V[2]–W[2],dashed); label("$\mathbf{v}_0$", V[0], NE); label("$\mathbf{v}_1$", V[1], NW); label("$\mathbf{v}_2$", V[2], NW); label("$\mathbf{w}_0$", W[0], S); label("$\mathbf{w}_1$", W[1], S); label("$\mathbf{w}_2$", W[2], S); [/asy] Then $\mathbf{v}$_n &= *proj*_{$\mathbf{v}$_0} $\mathbf{w}$_{n - 1}
&= $\frac{}{\{\mathbf{w}\_\{n-1\}\cdot\mathbf{v}\_0\}\{\|\mathbf{v}\_0\|^2\}\mathbf{v}\_0\&=\frac{\{}{b}\_\{n-1\}\cdot\mathbf{w}\_0\cdot\mathbf{v}\_0\}\{\|\mathbf{v}\_0\|^2\}\mathbf{v}\_0\&=\frac{\{}{b}\_\{n-1\}\binom{4}{0}\cdot\binom{1}{3}\}\{\left\|\binom{1}{3}\right\|^2\}\mathbf{v}\_0\&=\frac{2}{5}b\_\{n-1\}\mathbf{v}\_}$
$a_n = \frac{2}{5} b_{n-1}$. Similarly, $\mathbf{w}$_n &= *proj*_{$\mathbf{w}$_0} $\mathbf{v}$_n
&= $\frac{}{\{\mathbf{v}\_n\cdot\mathbf{w}\_0\}\{\|\mathbf{w}\_0\|^2\}\mathbf{w}\_0\&=\frac{\{}{a}\_n\cdot\mathbf{v}\_0\cdot\mathbf{w}\_0\}\{\|\mathbf{v}\_0\|^2\}\mathbf{w}\_0\&=\frac{\{}{a}\_n\binom{1}{3}\cdot\binom{4}{0}\}\{\left\|\binom{4}{0}\right\|^2\}\mathbf{w}\_0\&=\frac{1}{4}a\_n\mathbf{w}\_0.}$ Thus,
$b_n = \frac{1}{4} a_n$. Since $b_0 = 1$, $a_1 = \frac{2}{5}$. Also, for $n \geq 2$,

$$a_n = \frac{2}{5} b_{n-1} = \frac{2}{5} \cdot \frac{1}{4} a_{n-1} = \frac{1}{10} a_{n-1}.$$

Thus, (a_n) is a geometric sequence with first term $\frac{2}{5}$ and common ratio $\frac{1}{10}$, so $\mathbf{v}$_1 + $\mathbf{v}$_2 + $\mathbf{v}$_3 + &= $\frac{2}{5}$ $\mathbf{v_0}$ + $\frac{2}{5} \cdot \frac{1}{10}$ $\cdot$ $\mathbf{v}$_0 + $\frac{2}{5} \cdot$ ($\frac{1}{10}$)^2 $\cdot$ $\mathbf{v}$_0 +
&= $\frac{2/5}{1-1/10}$ $\mathbf{v}$_0 = $\frac{4}{9}$ $\mathbf{v}$_0 = {(4) /9
4/3}.
Note: Students should show all steps in their mathematical solution.

Answer: $32.93. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key